MARY C. BANNER

GIS & ENVIRONMENTAL SCIENCE

CARTOGRAPHY • REMOTE SENSING • ENVIRONMENTAL RESEARCH • SCIENCE COMMUNICATION



Yakima River | Cle Elum, Washington

Photograph by Mary C. Banner, Yakima River, Wa • Portfolio prepared for Nisqually Indian Tribe GIS Technician interview

Professional Profile & Portfolio Guide

PROFESSIONAL PROFILE

GIS and environmental science professional with experience in cartography, spatial data integration, environmental research, fieldwork, and science communication. My work includes NASA ABoVE regional outreach mapping, research on manoomin (wild rice), applied fisheries fieldwork, and water operator.

EDUCATION

M.P.S. Environmental Science — SUNY ESF (2025) • B.S. Native Environmental Science — Northwest Indian College (2022) • A.A. Sociology — Pierce College (2019)

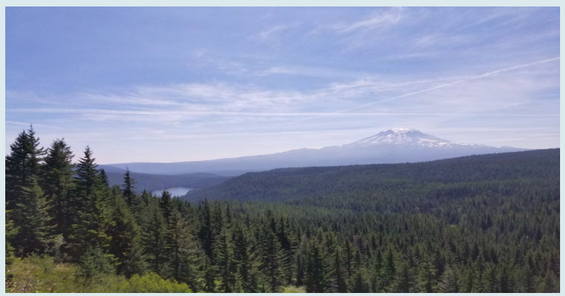
CAPABILITIES

ArcGIS Pro • QGIS • Cartographic layout • Georeferencing and digitization • Spatial data QA/QC • Environmental data interpretation • R analysis • Field data collection • Plain-language science communication

HOW TO READ THIS PORTFOLIO

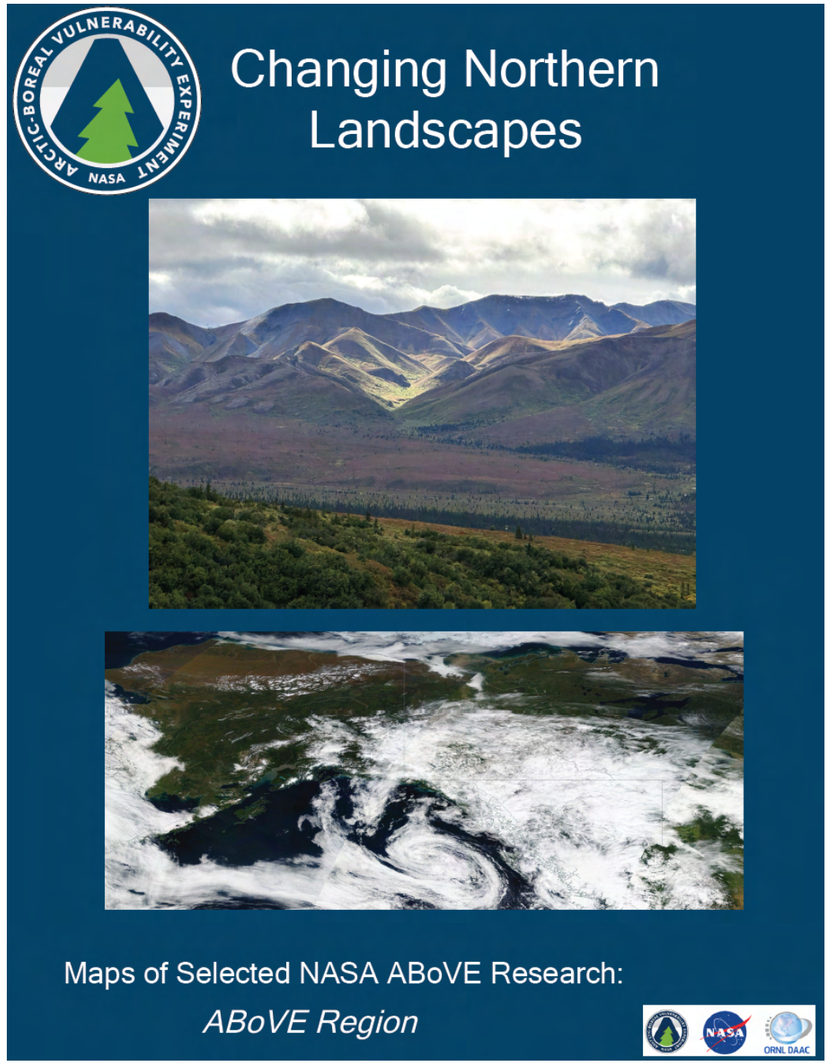
Selected cartography and science communication projects appear alongside research, fisheries fieldwork, and digital storytelling. Project descriptions distinguish my contributions from source datasets, photographs, and collaborative research.







Changing Northern Landscapes



NINE REGIONAL BOOKLETS

Changing Northern Landscapes communicated NASA ABoVE environmental research to northern communities through nine regional booklets, accessible maps, and plain-language explanations.

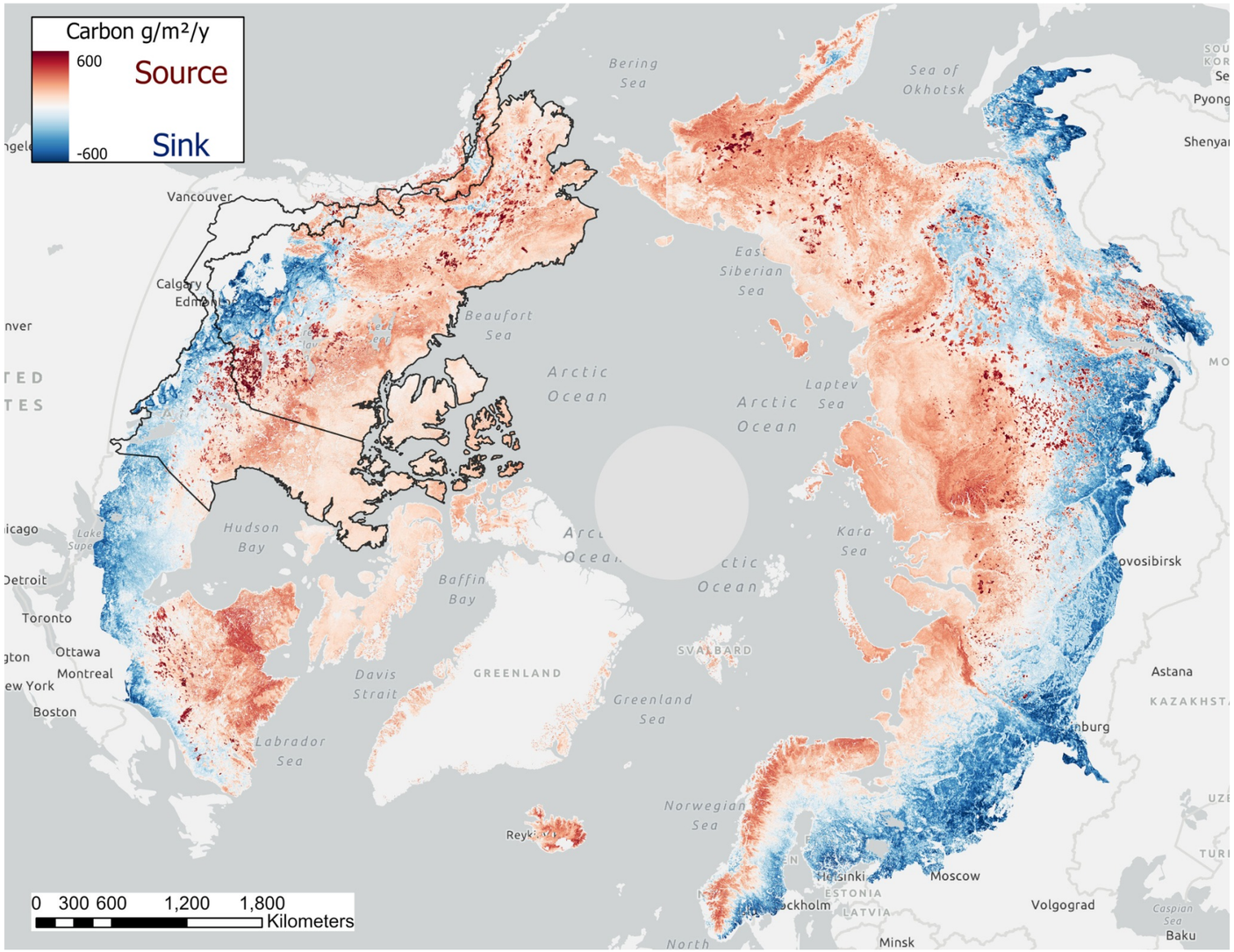
REGIONAL COVER APPROACH

Each regional edition used a landscape photograph together with satellite imagery to connect environmental change to recognizable places and scales.

MY CONTRIBUTIONS

Regional cartography, map production, data integration, QA/QC, outreach coordination, and plain-language editing. Research datasets and source photographs retain their original attribution.

Carbon Sources & Sinks



READING THE MAP

Warm colors indicate net carbon release (source); cool colors indicate net carbon uptake (sink).

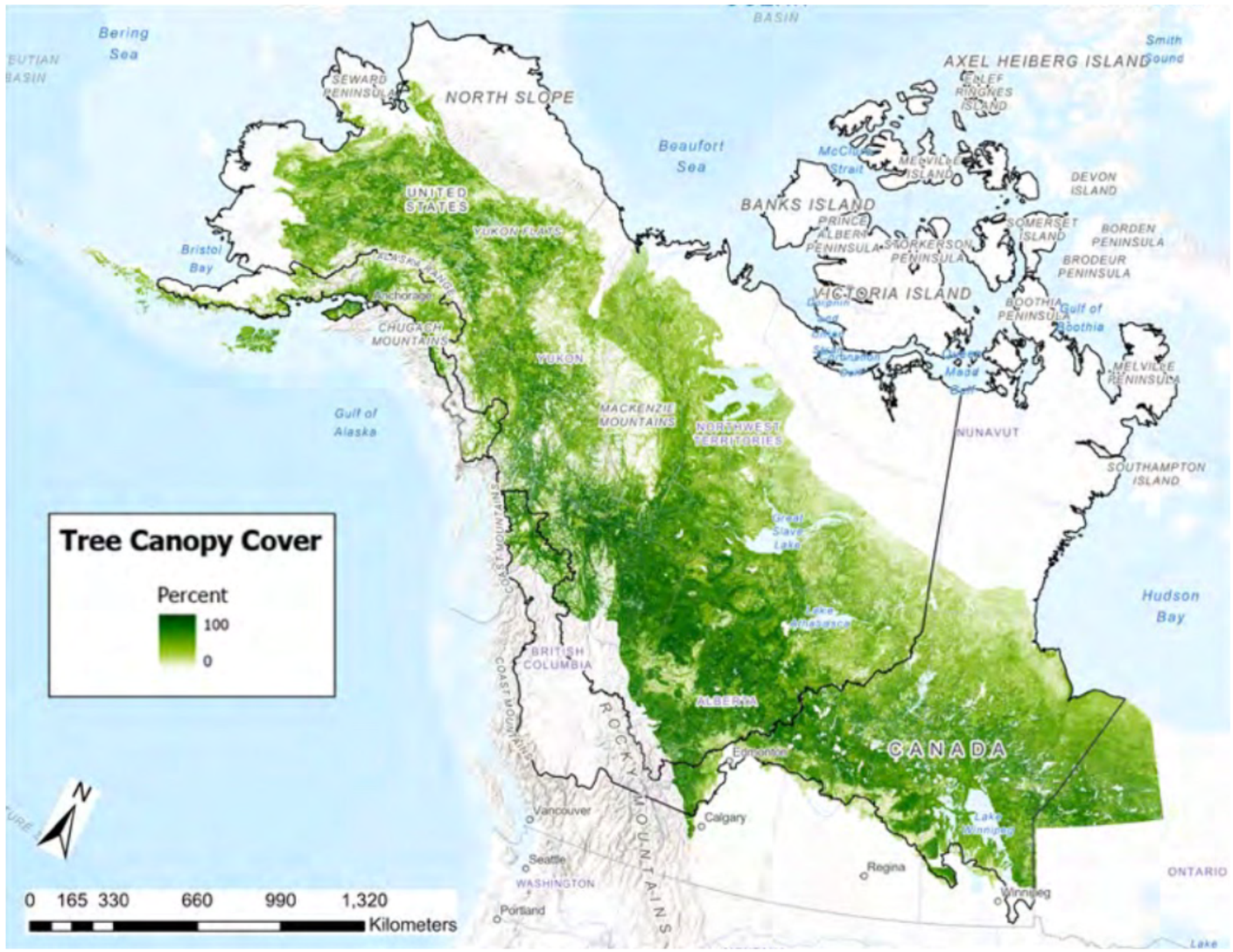
CONTEXT

Circumpolar carbon flux patterns illustrate how ecosystem processes vary across northern landscapes.

CREDIT

ABoVE research data visualization; scientific data attribution should remain with the original dataset.

Tree Canopy Cover



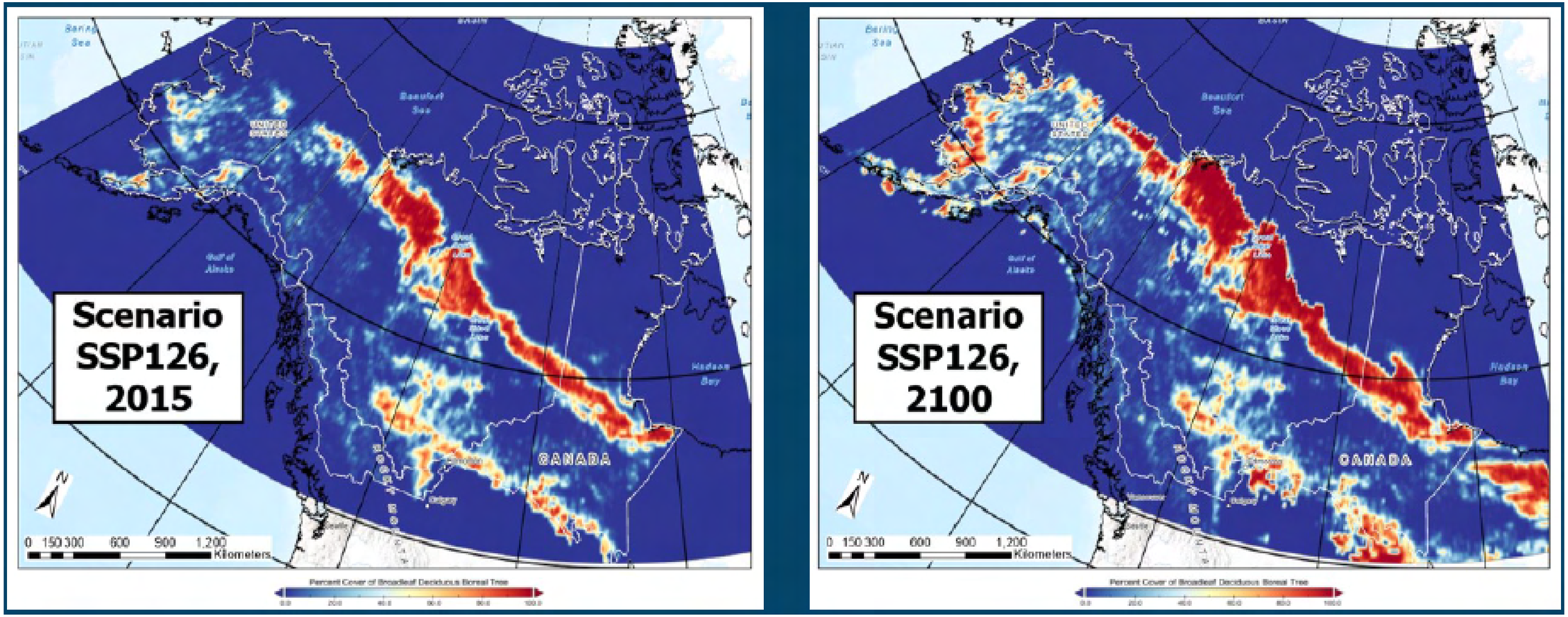
CANOPY COVER

The regional tree canopy map uses a green color ramp to communicate variation in canopy cover across the ABoVE region.

PORTFOLIO FOCUS

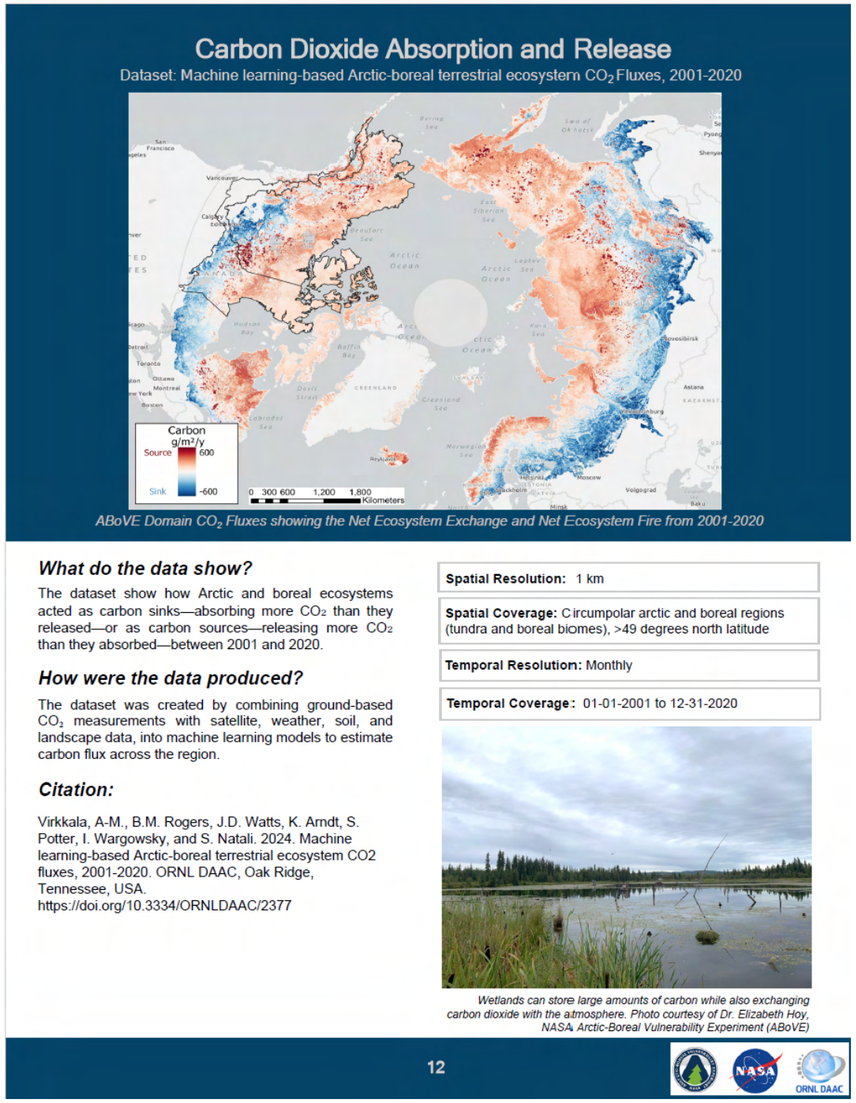
Map legibility, consistent legends, regional framing, and accessible scientific communication.

Landscape Change Scenarios



Comparison of the percentage cover of deciduous boreal trees under climate scenario SSP1-2.6 in 2015 and 2100. The side-by-side maps highlight projected changes in tree cover over time.

Inside the Booklet



Carbon Dioxide Absorption and Release

Dataset: Machine learning-based Arctic-boreal terrestrial ecosystem CO_2 Fluxes, 2001-2020

ABoVE Domain CO_2 Fluxes showing the Net Ecosystem Exchange and Net Ecosystem Fire from 2001-2020

What do the data show?

The dataset show how Arctic and boreal ecosystems acted as carbon sinks—absorbing more CO_2 than they released—or as carbon sources—releasing more CO_2 than they absorbed—between 2001 and 2020.

How were the data produced?

The dataset was created by combining ground-based CO_2 measurements with satellite, weather, soil, and landscape data, into machine learning models to estimate carbon flux across the region.

Citation:

Virkkala, A-M., B.M. Rogers, J.D. Watts, K. Arndt, S. Potter, I. Wargowsky, and S. Natali. 2024. Machine learning-based Arctic-boreal terrestrial ecosystem CO2 fluxes, 2001-2020. ORNL DAAC, Oak Ridge, Tennessee, USA. https://doi.org/10.3334/ORNLDAAC/2377

Spatial Resolution: 1 km

Spatial Coverage: Circumpolar arctic and boreal regions (tundra and boreal biomes), >49 degrees north latitude

Temporal Resolution: Monthly

Temporal Coverage: 01-01-2001 to 12-31-2020

Wetlands can store large amounts of carbon while also exchanging carbon dioxide with the atmosphere. Photo courtesy of Dr. Elizabeth Hoy, NASA Arctic-Boreal Vulnerability Experiment (ABoVE)

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ORNL DAAC

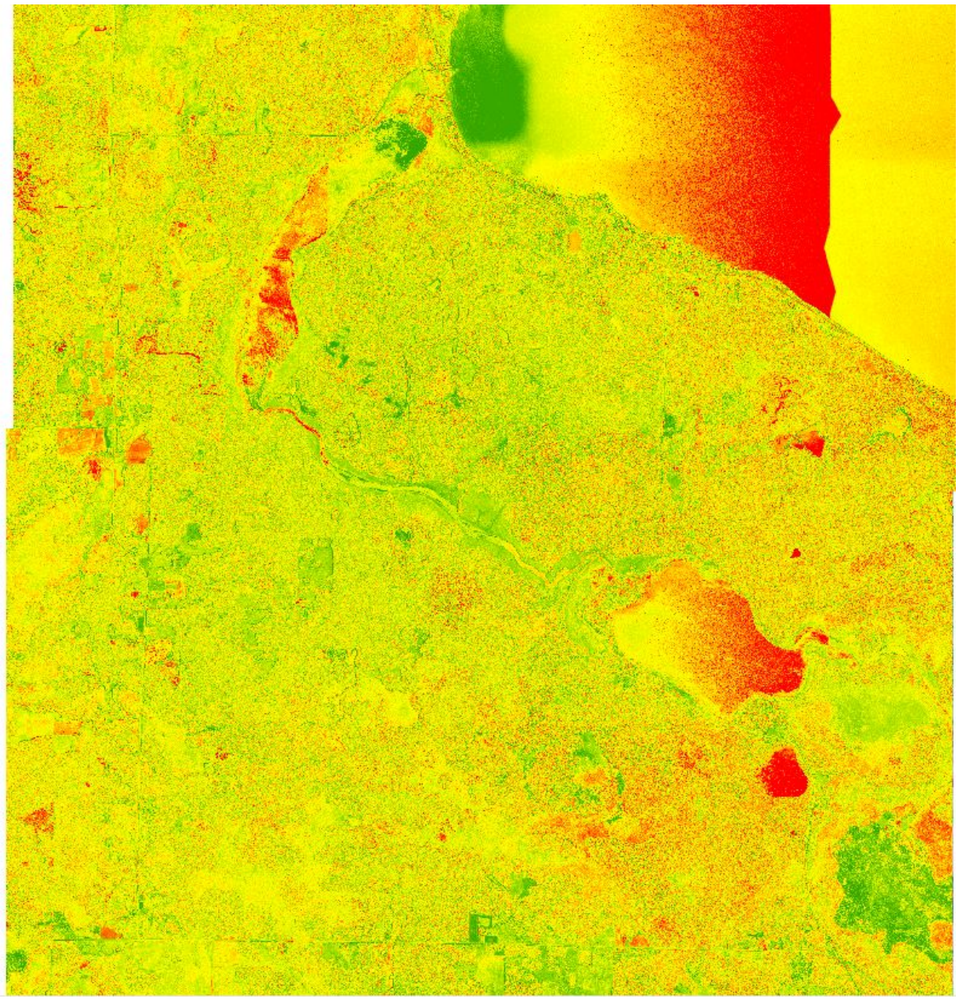
SCIENCE COMMUNICATION

The finished booklet page combines a carbon flux map with an explanation of the data, methods, resolution, temporal coverage, citations, and supporting photography.

PHOTO ATTRIBUTION

Wetland photograph credited to Dr. Elizabeth Hoy, NASA ABoVE, as shown in the original booklet caption.

Manoomin | Manoomin | Pre-Construction Change (2008–2014)



2008–2014 Before Construction of Dam

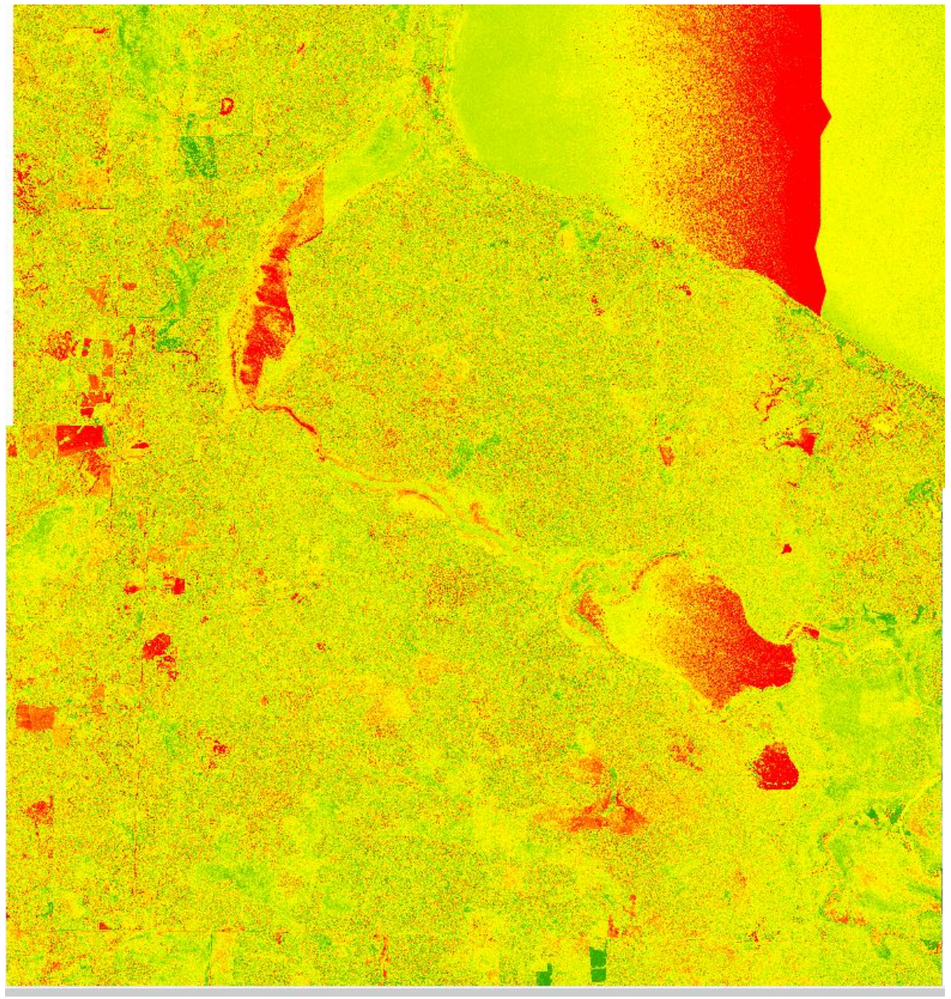
Using USDA NAIP aerial imagery and ArcMap s Image Differencing Change Detection workflow, I examined changes in manoomin (wild rice) beds near gechie ake, Rum River, and Shakopee ake, Minnesota, before the 2014 Buckmore Dam construction efforts.

The analysis compared imagery from 2008 and 2014 to identify changes in vegetation and surrounding landscapes. Red indicates greater change, yellow indicates minor change, and green represents areas of relatively little change.

Data & Software: USDA NAIP imagery, USGS EarthExplorer, ArcMap.

Map: Mary C. Banner | Comparison period: 2008–2014

Manoomin | Post-Construction Change (2014–2017)



2014–2017 After Construction Efforts

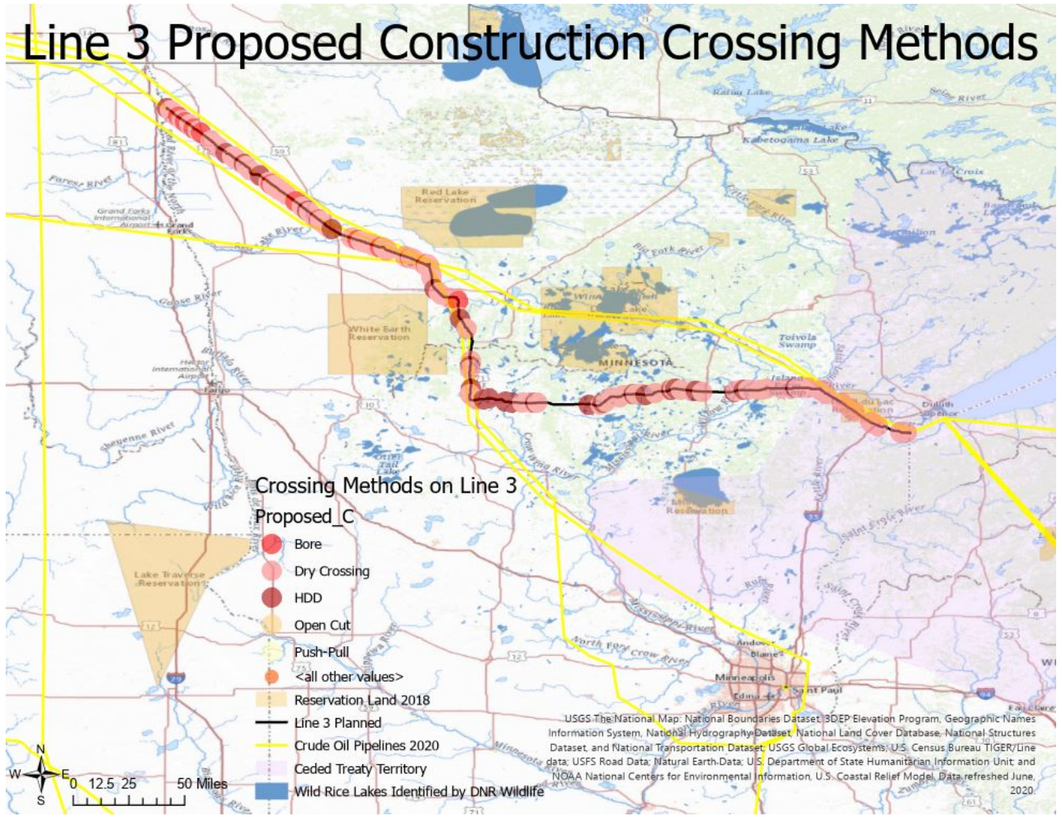
Using the same Image Differencing Change Detection workflow, I compared USDA NAIP aerial imagery from 2014 and 2017 to investigate changes in manoomin beds following construction activities associated with the Buckmore Dam proposal.

Compared with the earlier period, differences in vegetation patterns and color intensity are visible, including changes in green tones around the study area. These results highlight areas of potential vegetation change that warrant further investigation into impacts on manoomin habitat.

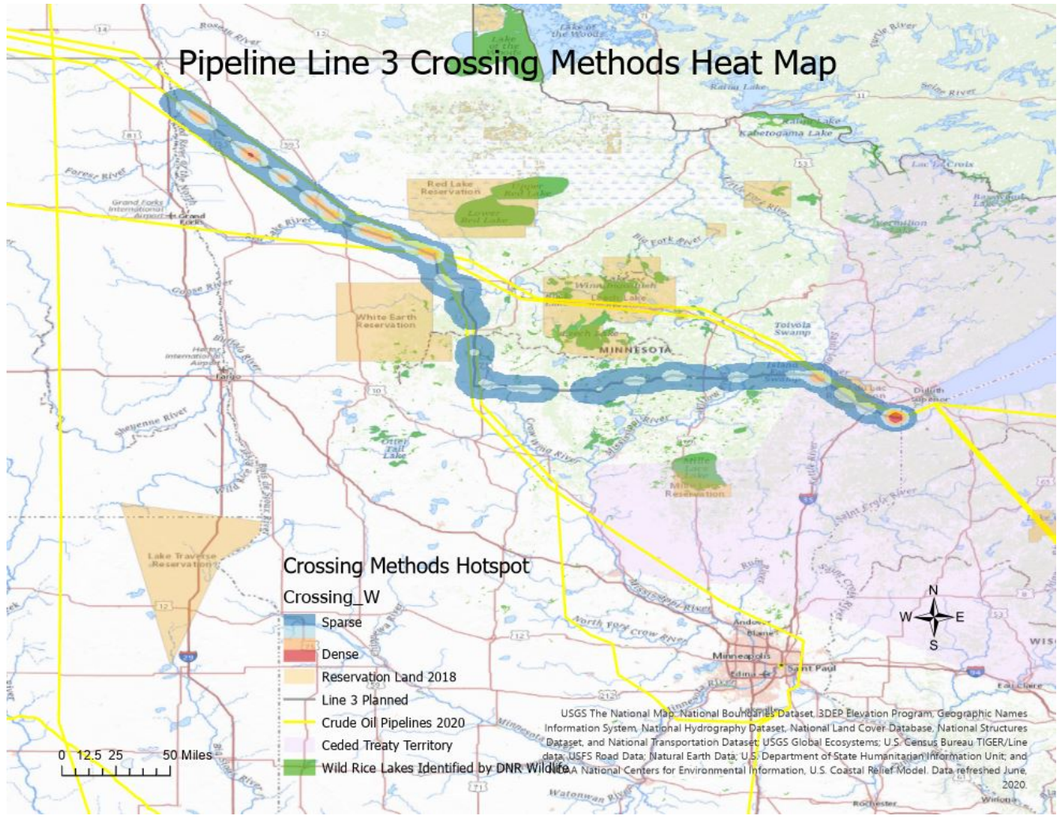
Data & Software: USDA NAIP imagery, USGS EarthExplorer, ArcMap

Map: Mary C. Banner | Comparison period: 2014–2017

Enbridge Line 3 | Regional GIS Maps



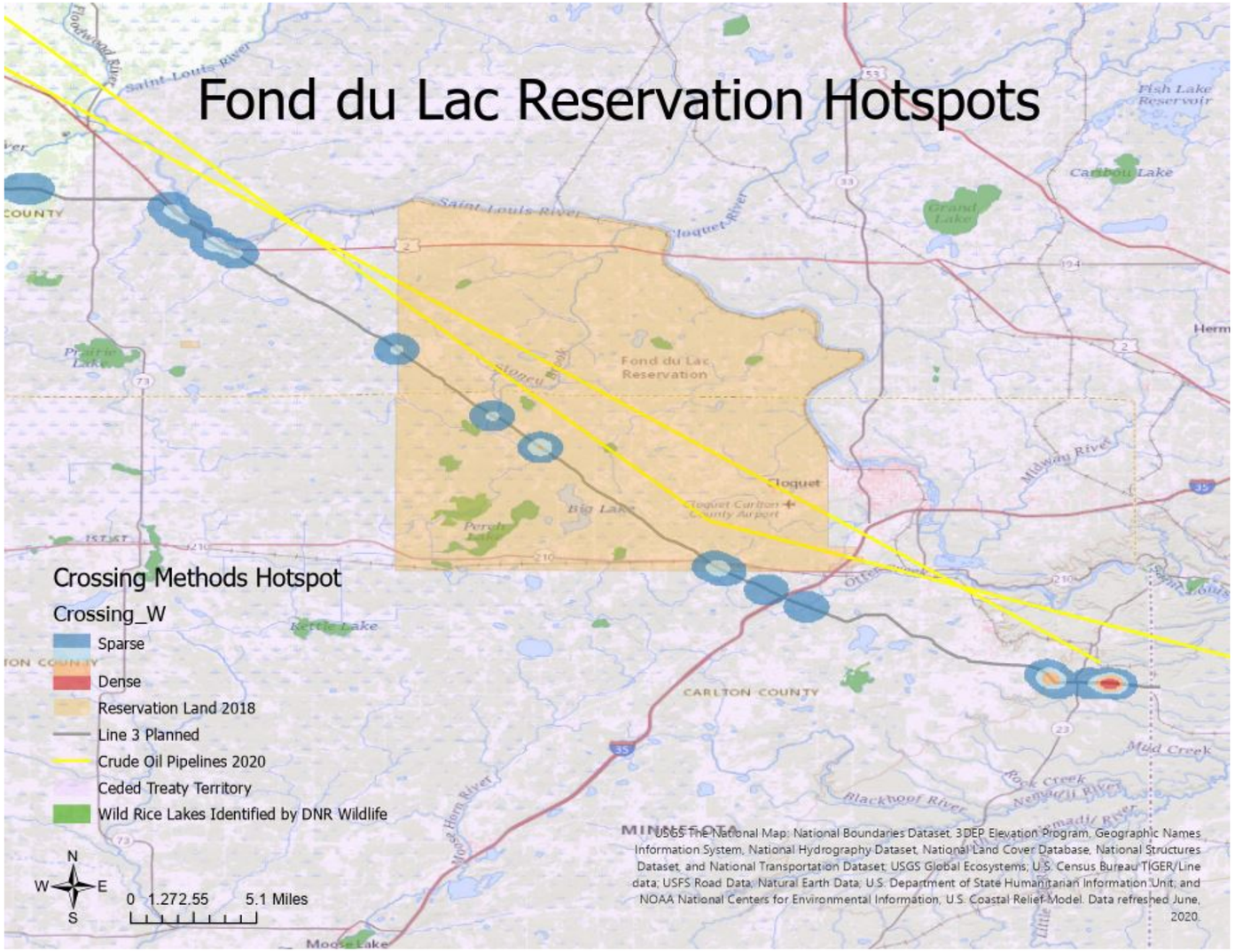
Proposed construction crossing methods



Crossing-method concentration / hotspot view

Primary Salish Sea Research Center mapping products examining the Enbridge Line 3 corridor, construction crossings methods, and geographic concentrations.

Fond du Lac Reservation | Hotspots



MAPPING FOCUS

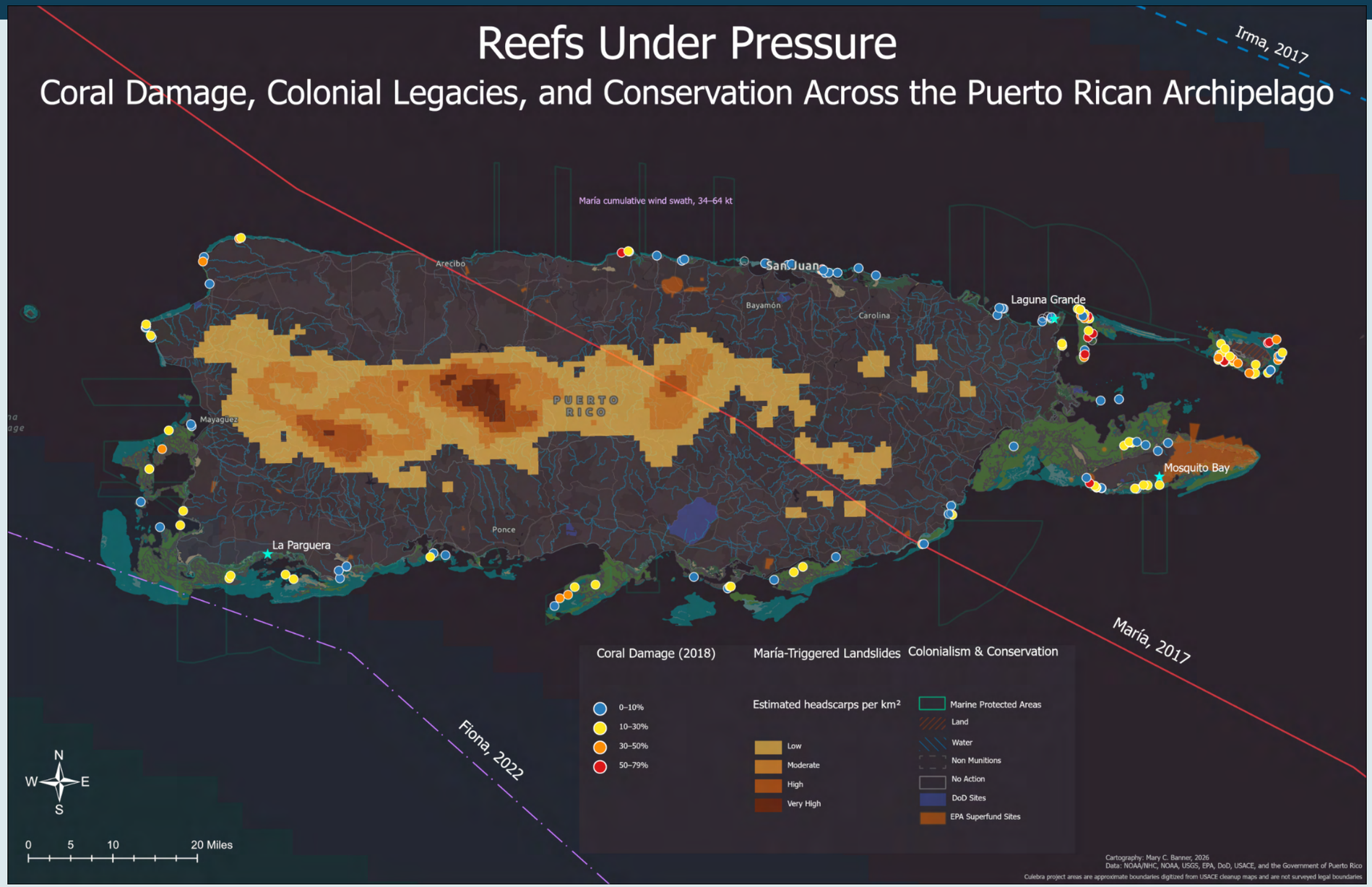
Detailed hotspot analysis in relation to the Fond du Lac Reservation and Enbridge Line 3 corridor.

MY CONTRIBUTION

Original GIS cartography and spatial interpretation developed through Salish Sea Research Center research.

Reefs Under Pressure

Coral damage, colonial legacies, and conservation across the Puerto Rican archipelago. Cartography: Mary C. Banner, 2026.



Reefs Under Pressure | Project Overview

Coral damage, colonial legacies, and conservation across the Puerto Rican archipelago

Project Overview

This independent cartographic project examines overlapping environmental pressures across Puerto Rico, Vieques, and Culebra. I integrated hurricane impacts, coral reef observations, marine habitats, protected areas, and historical military land-use data to explore how natural disturbances and human activities intersect across the archipelago.

GIS Methods & Data

Using ArcGIS Pro, I combined hurricane tracks and wind data for Irma (2017), María (2017), and Fiona (2022) with coral damage observations, benthic habitats, and marine conservation boundaries. I also included EPA Superfund and Department of Defense sites, georeferenced Culebra cleanup maps, and digitized approximate former military-area boundaries.

Environmental Significance

The map brings together hurricane-related disturbances, reef vulnerability, conservation areas, and historical military activity. These spatial relationships support broader environmental questions without asserting that a particular site caused observed coral damage.

Technical Skills Demonstrated

ArcGIS Pro • Spatial data integration • Georeferencing • Digitization • Cartographic layout • Environmental data visualization • Spatial interpretation

NASA Indigenous Peoples Initiative

TRIBAL MIDWEST EARTH OBSERVATION SUMMARY

NASA Indigenous Peoples Initiative | Summer 2023

I researched how NASA Earth observation (EO) data and geospatial tools could support Indigenous communities in the Great Lakes region. The white paper examined Tribal environmental programs, GIS capacity, and opportunities to connect satellite observations with community priorities and on-the-ground knowledge.

DELIVERABLE 20-page research white paper documenting findings and potential applications

RESEARCH SCOPE

Tribal environmental and GIS programs in Minnesota, Wisconsin, Michigan, and Iowa; Tribal colleges and geospatial education.

ENVIRONMENTAL FOCUS

Water quality, manoomin, fisheries, habitat restoration, invasive species, forestry, and climate adaptation.

PURPOSE

Identify ways NASA EO datasets and Tribal observations could inform monitoring, stewardship, and geospatial capacity building.

Earth Observations & Tribal Priorities

MANOOMIN & WETLANDS

The paper examines wild rice restoration and invasive species concerns, including opportunities to use Landsat and Sentinel-2 imagery to observe habitat and environmental change.

FISHERIES & WATER

Examples include Tribal fish stocking, water quality monitoring, and potential applications of remotely sensed temperature, chlorophyll, and watershed information.

FORESTS & HABITATS

Research explores land-cover change, vegetation indices, drones, prescribed fire, wildlife habitat, and monitoring invasive species with EO data.

RESEARCH OUTCOME

The white paper identifies opportunities to combine NASA satellite data with existing Tribal environmental programs, local observations, and Indigenous knowledge. It emphasizes community-specific tools, relationships, training, and capacity building rather than a single solution for every Tribal Nation.

SOURCE / Mary C. Banner, Tribal Midwest EO Summary, NASA Indigenous Peoples Initiative (2023)

Yakama Nation Fisheries | Fieldwork



Sculpin measurement during fisheries fieldwork



Priest Rapids Dam • Sockee handling and transport to Lake Cle Elum, WA

Field experience included fish measurement, handling operations, and transporting fish for release or relocation. Photographs by Mary C. Banner.

Yakama Nation Fisheries | Conservation



Searching for bull trout stranded in drying pools



Water-control equipment during field operations

Fisheries crews searched beneath large woody debris for bull trout in temporary pools that were expected to dry overnight. Photographs document field conditions and conservation activities.

Water System Operations

Fox Island Mutual Water Association | Operator in Training | July 2025 – April 2026



01 DISTRIBUTION LINE FLUSHING

Flushing equipment connected at a system access point.



02 HYDRANT FLUSHING

Hydrant and hose setup for distribution system flushing.



03 BLOW-OFF INSPECTION

Field inspection of a blow-off assembly and access point.

OPERATIONAL EXPERIENCE

Supported routine maintenance and field operations for a community drinking-water distribution system, including line and hydrant flushing, valve and blow-off exercising and inspections, meter readings, water quality sampling, leak repairs, and emergency response. This fieldwork developed practical familiarity with utility infrastructure, maintenance procedures, and accurate system documentation.

Additional Work & Contact

INTERACTIVE STORYMAPS

1. https://arcg.is/1jGavm2 - Phylum Cnidaria
2. https://arcg.is/qDn5q0 - Communicating Complex Science Simply
3. https://arcg.is/1q1XHr - The Glow of Pyrodinium bahamense
4. https://arcg.is/1v9ey02 - Effects of Anthropogenic Activities on Manoomin (Wild Rice)

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CONTACT

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